STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
CAT 1
DSA 8301: Statistical Inference for Big Data

DATE: 8th April 2026 **TIME:1hr 30 Min**

Instruction:

Answer All Question

1. Let $X_1, X_2, \ldots, X_n \sim f(x \mid \theta)$, where $\theta \in \Theta \subset \mathbb{R}^p$.

   (a) Define a **parametric statistical model**. Clearly describe $\mathcal{X}$ (**Sample Space**), Θ (**Parameter Space**), and $\mathcal{M}$ (**Model / Family of Distributions**)
   . **(4 marks)**

   (b) Explain why finite-dimensional parameter spaces are important in statistical inference and big data applications
   . **(2 marks)**

   (c) Consider

   $$f(x|\theta) = \frac{1}{\theta}e^{-x/\theta}, \quad x > 0.$$

   (i) Identify the distribution. **(1 mark)**

   (ii) Specify $\mathcal{X}$ and Θ. **(2 marks)**

   (d) Discuss two limitations of parametric models. **(2 marks)**

2. Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with mean μ.

   (a) Define a point estimator. **(1 mark)**

   (b) Show that the sample mean $\bar{X}$ is an unbiased estimator of μ. **(3 mark)**

   (c) Given the sample values

   $$4, 6, 5, 7, 3, 5, 6, 4,$$

   compute the point estimate of μ. **(1 mark)**

3. (a) Let

   $$\bar{X} = \frac{1}{n}\sum X_i.$$

(i) Find $\mathbb{E}[\bar{X}]$ and $\mathrm{Var}(\bar{X})$. **(3 marks)**

(ii) Explain the role of the CLT. **(2 marks)**

(b) Discuss the effect of large sample sizes on inference. **(2 marks)**

4. (a) Define consistency of an estimator. **[2 marks]**

(b) State and prove a sufficient condition for consistency using bias and variance
. **[5 marks]**

(c) Using Chebyshev's inequality, show that

$$\mathrm{MSE}(\hat{\theta}_n) \to 0 \implies \hat{\theta}_n \xrightarrow{p} \theta.$$

[4 marks]

(d) Show that $\bar{X}_n$ is a consistent estimator of μ. **[3 marks]**